Industrial Safety Processes

Working at Heights

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Canada Labour Code Part II – 12.07

12.07 (1) Subject to subsection (2), an employer must provide or put in place a fall-protection system if work is to be performed

a) from a structure or on a vehicle at a height of 3 m or more;

b) from a ladder at a height of 3 m or more if, because of the nature of the work, the person performing it is unable to use at least one hand to hold onto the ladder; or

c) at a height of less than 3 m if the surface onto which the person might fall would present a greater risk of injury than a solid, flat surface.

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Canada Labour Code Part II – 12.07

(2) If an employee is required to work on a vehicle and it is not feasible to provide or put in place a fall-protection system, an employer must

a) in consultation with the policy committee or, if there is no policy committee, the work place committee or the health and safety representative,

   i. perform a job safety analysis to eliminate or minimize the need for the employee to climb onto the vehicle or its load, and

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Canada Labour Code Part II – 12.07

ii. provide every employee who could be required to climb onto the vehicle or its load with instruction and training by a qualified person on the safe method of climbing onto it and working there;

b) make a report in writing to the Minister setting out the reasons why it is not feasible to provide or put in place a fall-protection system and include in that report the job safety analysis and a description of the instruction and training referred to in paragraph (a); and

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Canada Labour Code Part II – 12.07

c) provide a copy of the report referred to in paragraph (b) to the policy committee or, if there is no policy committee, the work place committee or the health and safety representative.

(3) The job safety analysis, instruction and training referred to in paragraph (2)(a) must be reviewed every two years in consultation with the policy committee or, if there is no policy committee, the work place committee or the health and safety representative.

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Temporary Work Areas

- Some provincial jurisdictions separate “temporary” work areas from “permanent” work areas
- This permits some flexibility in the application of fall protection controls
- In these situations the implementation of a formal hazard assessment program (job safety analysis in the CLC Part 2) and site fall protection plans are standard

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Canada Labour Code Part II – 12.01

fall-protection system means a system that is designed and configured to eliminate or reduce the risk of a person falling, restrain a person who is at risk of falling or arrest a person's fall. The system may be composed of one or more of the following:

a) a passive fall-protection system;

b) a fall-restraint system;

c) a fall-arrest system;

d) a fall hazard zone system.

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Canada Labour Code Part II – 12.01

- ***passive fall-protection system*** means a system of physical barriers that is designed and installed to prevent a person from falling, including guardrails, fences, barricades or covers.

- ***fall-restraint system*** means a collection of protection equipment that attaches a person to an anchorage and is designed and configured to prevent the person from getting close to an unguarded edge.

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Canada Labour Code Part II – 12.01

- ***fall-arrest system*** means a collection of protection equipment that attaches a person to an anchorage and is designed and configured to arrest a free fall.

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Types of Arrest Systems

- Personal fall protection devices
  - Lanyards, shock absorbing lanyards
  - Body harnesses
- Ladder Cages
  - Height requirements specified
- Safety Nets
  - Reliance on Workers' Compensation Board is not a "safety net"

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Canada Labour Code Part II – 12.01

- ***personal fall-protection system*** means a fall-restraint system or fall-arrest system.

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Canada Labour Code Part II – 12.01

- ***fall hazard zone*** means a defined area bordering on an edge that is unguarded against a risk of falling.
- ***Fall hazard zone system*** means a system that consists of a fall hazard zone that is supervised by a fall hazard zone monitor.

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Canada Labour Code Part II – 12.08

12.08 (1) A fall hazard zone must be at least 2 m wide and extend the entire length of the unguarded edge.

2) A fall hazard zone must only be established on a surface that has a slope of five degrees or less.

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Canada Labour Code Part II – 12.08

(3) If a person needs to carry out an activity in a fall hazard zone or cross a fall hazard zone to get to or from a work area,

a) the employer must ensure the presence of a fall hazard zone monitor whose exclusive duties are to

i. supervise the fall hazard zone whenever an activity is being carried out there, and

ii. ensure that the requirements of the fall-protection plan are respected;

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b) the employer must install, along the border between the fall hazard zone and any other work area, a raised demarcation line at a height of not less than 900 mm but not more than 1 100 mm; and

c) the employer must ensure that a personal fall-protection system is used.

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Canada Labour Code Part II – 12.09

(8) The employer must ensure that a person who works on an aerial device, boom-type elevating platform, scissor lift platform, forklift truck platform or any similar personnel lifting equipment in the circumstances described in subsection 12.07(1) uses a fall-restraint system that is connected to

a) an anchorage that is specified in the instructions of the manufacturer of the lifting equipment; or

b) if no anchorage is specified by the manufacturer, an anchorage that is certified by a person who is authorized to exercise the profession of engineering in Canada and that meets the requirements set out in CSA Group Standard Z259.16, Design of active fall protectionsystems.

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Man Baskets

- Scissor lifts
- Zoom booms
- Attachments to forklifts or other equipment

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Provincial vs. Federal

- The Canada Labour Code has recently been updated to more closely match various provincial regulations
- There are still some potential conflicts – so know the legislation in your area
  - Work zones, adopted standards, rescue standards, requirements for mobile elevated work platforms

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Key Things to Know

- At what height are you “working at heights” and require fall protection?
- What is specifically required for “unusual” situations – work zones / control zones, temporary access?
- What do clients / Prime Contractors / Constructors / etc. require?

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Personal Fall Arrest Hardware

- Three main components:
  - Harness
  - Lanyard
  - Anchor Point
- All three must be present
- Don’t confuse with non-personal fall arrest systems – i.e., netting

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Canada Labour Code Part II – 12.09

(6) The employer must ensure that a person who is using a personal fall-protection system wears and uses a full body harness.

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Harnesses

- Full body harnesses are the only holding devices approved for fall arrest systems
- Maximum capacity is typically 310 lbs. – check specifications
- Harnesses must be compliant with current regulations and their use must conform to class of harness

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Harness Classes

There are 5 different classes of harnesses:

- Group A – Fall Arresting
- Group D – Controlled Descent
- Group E – Confined Space Entry
- Group L – Ladder Climbing
- Group P – Work Positioning

L – E – A – P - eD

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Identifying Harnesses

- AEL
  - Fall Arrest, Confined Space, Ladder Climbing
- ELP
  - Confined Space, Ladder Climbing, Work Positioning
- Use only the harness class rated for the intended activity

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Lanyards

- These are the lines between the worker and the anchor point
- Lanyards come in several variations and sizes
  - Rope
  - Web
  - Wire Rope (cable)
  - Retractable (SRLs)
  - 3 feet to whatever length

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Using Lanyards

- Use the appropriate lanyard for the hazard and work being performed
  - Welder – wire rope, not web
- Ensure worker understands how lanyard works, how to use it, and what to do if there is a problem with the unit
- Ensure manufacturer's specifications, requirements, and limitations are followed

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Connecting Devices

- D-Rings, O-Rings, Carabiners, Hooks
- Avoid stresses on wrong areas
- Devices must be self-closing and self locking
- Devices must require 2-action manipulation to avoid accidental release

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Anchor Points

- Point of attachment for worker
- Required for permanent and temporary work which requires personal fall arrest or restraint systems
- Specifications for different applications are given in the different provincial regulations
- May be permanently anchored or temporary
- 1 kilonewton – amount of force required to move one kilogram one metre in one second

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Lifelines

- Come in many varieties for different applications
  - Vertical
  - Horizontal
  - Travel Restraint Systems
  - Fall Arrest Systems (ladders, SLRs, scaffolding)

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Canada Labour Code Part II – 12.06

12.06 (1) If there is a risk of injury due to falling in any of the circumstances described in paragraphs 12.07(1)(a), (b) or (c) in a work place, the employer must, before any work activities begin,

a) develop a fall-protection plan in consultation with the work place committee or the health and safety representative; and

b) ensure that a copy of the fall-protection plan is readily available at the work place for consultation.

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Canada Labour Code Part II – 12.06

(2) The fall-protection plan must specify

a) the hazards that have been identified for each work area and each activity to be carried out at the work place;

b) the fall-protection systems that have been chosen to protect against the identified hazards;

c) if a personal fall-protection system is used, the anchorage to be used during the work;

d) if a fall-arrest system is used, the clearance distance below each work area;

e) if there are no manufacturer's instructions with respect to the storage, maintenance, inspection, testing, fitting, installation, use or dismantling of equipment that is used in a fall-protection system and that is provided by an employer, the procedures to be followed for the purposes of paragraphs 12.05(1)(a) to (d); and

f) the rescue procedures to be followed if a person falls.

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Canada Labour Code Part II – 12.06

(3) The fall-protection plan must include, in annex, a copy of any manufacturer's instructions with respect to the storage, maintenance, inspection, testing, fitting, installation, use or dismantling of equipment that is used in a fall-protection system and that is provided by an employer.

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Key Items

- Identification of potential working at heights situations
  - Some provincial legislation excludes situations where guardrails or other engineering controls in place
- Systems to be used, including MOM's instructions
- Clearances in fall situations
- Rescue plans

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Application

- CLC Part II says a plan is required for “each work area”
- Some provincial legislation permits general plans for regular activity for mobile operations (i.e. utility repair, oil and gas rigs)
- Workers need to be trained in the plan and it has to be readily available

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Application

- Identify anchor points that meet or exceed regulatory requirements (force strength)
- Fall clearances from anchor points – ideally with specifics on permissible and / or non-permissible equipment
  - Fall restraint vs. arrest, length of lanyards, etc.

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Instruction of Workers – General

- Applicable legislation
- Fall protection plan, including work site hazards and rescue plan
- Equipment to used, including instructions on safe use, inspection, storage, maintenance and removal from service
- Impact on body from falling – arresting forces
- Practice in using equipment and implementing rescue plans

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Alberta OH&S Code Explanation Guide

In the case of rescues involving workers in confined spaces and workers suspended in the air after a fall, calling 911 alone and awaiting the arrival of rescue services personnel is considered to be an insufficient emergency response. The employer must have some basic level of on-site rescue capability ….in the event that rescue services personnel are delayed or unable to attend the scene.

Government of Alberta, Occupational Health and Safety Code 2018 Explanation Guide, p. 9-11.

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Incompetence? Intolerance?

- Orthostatic intolerance or incompetence
- May kill a worker hanging in a harness in as little as 5 minutes
- Quick rescue is critical – see "Will Your Safety Harness Kill You?"
  - http://www.elcosh.org/en/document/590/d000568/will-your-safety-harness-kill-you%253F.html

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- Legs in static position do not pump blood back to heart – muscles pressing veins, one-way valves, etc.
- Blood begins to pool in legs, reducing blood to heart and brain (just like safety classes)
- Heart tries to compensate by beating faster – more blood pools in legs
- Body slows heart – lose consciousness

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Problem in Suspension

- On your own feet, you fall down and blood returns to head and heart as you lay on ground
  - Soldiers fainting on parade ground
- In harness, you don’t fall down – blood does not return to head and heart

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Major Problem for Rescue

- Response for rescue must be quick
- May require alarm system, 2-man teams, etc.
- Cannot rely on fire department
- May need special equipment and training
- Organization and workers need to examine options & alternatives

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After Rescue – He Died?

- Lowered person may still die if laid down too soon
- Blood rushes back too fast to heart
- May need to take as much as 30-40 minutes to move person from kneeling to sitting to lying position
- Person needs to be examined by a doctor ASAP

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